

Kaiming Liu

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Research Profile

Scientific machine learning for magnetic inverse problems: probabilistic models, physical forward validation, and simulation-to-experiment robustness. Background in experimental condensed matter and computational many-body physics.

Education

University of Illinois Urbana-Champaign

Ph.D. in Physics, in progress

Aug 2023–present

Advisor: Bryan K. Clark

Xi'an Jiaotong University

B.S. in Physics (Honors)

Sep 2019–Jul 2023

Special Class for the Gifted Young

Sep 2017–Jul 2019

University of California, Berkeley

BPIE Program (non-degree visiting program)

Aug–Dec 2021

Current Research

University of Illinois Urbana-Champaign

Graduate researcher, Bryan K. Clark group

Nov 2025–present

Crystal-field inverse modeling

- Infer multiple crystal-field candidates from susceptibility with a residual MLP and conditional Gaussian mixture models (cGMMs).
- Check candidates with forward susceptibility simulation; test noise augmentation on held-out synthetic curves and experimental measurements.

Magnetic-scattering inverse modeling

- Infer exchange parameters from powder-averaged neutron-scattering maps using MLP and ResNet conditional Gaussian mixtures.
- Use exact Sunny forward checks and classical fits as references; test sensitivity to noise, intensity scale, and background offsets.

Neural-network backflow / variational Monte Carlo

- Implement JAX neural-network backflow wavefunctions and variational Monte Carlo for lattice Hubbard models.
- Profile parallel GPU sampling, local-energy evaluation, and gradient synchronization; check chunked evaluation against the original computation.

Publications	[P1] Evidence of intertwined pair density and charge density wave orders in UTe₂ Zhen Zhu, Yudi Huang, Julian May-Mann, Kaiming Liu, Zheyu Wu, Shanta R. Saha, Johnpierre Paglione, Alexander G. Eaton, Andrej Cabala, Michal Vališka, Eduardo Fradkin, Vidya Madhavan. <i>PNAS</i> 123 (29), e2602117123 (2026). arXiv.	
Prior Research	University of Illinois Urbana-Champaign <i>Research assistant, Vidya Madhavan group</i> <i>Aug 2023–Oct 2025</i> • Grew thin-film heterostructures with MBE and characterized electronic structure using STM/STS in ultrahigh vacuum. • Automated growth temperature control and built Python/MATLAB analysis of STM/STS quasiparticle-interference data. Xi'an Jiaotong University <i>Research assistant, Yongchang Zhang group</i> <i>Mar 2022–Jul 2023</i> • Simulated Rydberg-atom dynamics and photon-mediated interactions in Python/MATLAB to study quantum nonlinear optics. • Built a discrete-time lattice model of two-qubit phase-flip dynamics and many-body regimes. University of California, Berkeley <i>Research assistant, Michael F. Crommie group</i> <i>Oct–Dec 2021</i> • Assisted synthesis and STM characterization of graphene nanoribbons with five-membered rings to study their electronic states.	
Teaching	University of Illinois Urbana-Champaign <i>Teaching assistant, PHYS 212</i> <i>Spring & Fall 2026</i> University of Illinois Urbana-Champaign <i>Teaching assistant, PHYS 102</i> <i>Aug–Dec 2023</i>	
Selected Honors	China National Scholarship 2021 Xi'an Jiaotong University Mathematical Contest in Modeling, Meritorious Winner 2021 COMAP	
Technical Skills	<i>ML / scientific computing:</i> Python (PyTorch, JAX, NumPy, SciPy), MATLAB; conditional mixture models; Git, Linux. <i>Computational physics / HPC:</i> MCMC, variational Monte Carlo, numerical optimization; GPU/cluster computation and runtime profiling. <i>Experiment:</i> MBE, STM/STS, ultrahigh vacuum, feedback control, experimental automation and data analysis.	